

Keep it Moving: Crop Rotation

Denyse Cummins, MS

Louisiana Farm to School Program Advisor



**GROWING NEW
CONNECTIONS**

2023 Louisiana Farm to School Conference

Louisiana State Government Supports School Gardens

- Act 404 (2016) requires BESE and LDAF to develop a farm to school program
- Seeds to Success: The Louisiana Farm to School Program
- School Gardens are incorporated into the State Pesticide Law



SEEDS to SUCCESS
THE LOUISIANA FARM TO SCHOOL PROGRAM



Provisions of the Law

(Pesticides in Schools, Part VIII)

3381, 3382 Purpose & definitions

3383 School shall keep a record of pesticides

3384 8 hours must pass before possible exposure

3385 Use least toxic method of control possible

3386 Applied by a certified commercial applicator

3387 Schools must employ a commercial applicator

3388 Have an annual IPM plan

3389 Schools maintain a registry of hypersensitive students

<https://www.beyondpesticides.org/assets/media/documents/documents/LASchoolLaw.pdf>



How Does IPM Work?

- IPM – Integrated Pest Management
 1. Brings together many systems that work together
 2. Scout for problems. Decide if there is enough damage to warrant taking action
 3. Exhaust all cultural practices (mulching, rotation, sanitation, variety selection)
 4. Chemical control is always the last resort
- Cultural practices are a major part of IPM



What is a 'Cultural Practice'?

- Cultural practices are those things we always do in the garden without thinking
 - Weeding, watering, fertilizing
 - Deeply ingrained, we know we have to do them to have a productive garden
- Some cultural practices require a little more thought and planning
 - Mulching
 - Providing plant supports
 - Planting the right crop in the right season
 - Crop rotation



The Easiest Cultural Practice to Avoid Problems

- Variety Selection
 - Plant breeding has produced varieties with resistance to some of the worst diseases
 - V = Verticillium wilt
 - F = Fusarium wilt
 - N = Nematode resistance
 - Corn bred with tight husks to exclude corn ear worm
 - Varieties with resistance to rots & foliar diseases
- Check your Sustainable Growing Guides for resistant cv's



Table 1. Recommended Beet and Carrot Varieties for Louisiana

Variety Name	Description	Days to Harvest & Size	Resistance
Beet			
Bull's Blood	Deep burgundy leaves and roots; candy-striped roots when sliced; great for beet greens; flavorful and tender as baby beet; tolerates low light, heat, and cold; French heirloom variety	58-60 days (35 days for baby leaf)	
Chioggia	Green tops; medium height; scarlet-pink striped roots; very flavorful; early maturing and vigorous; good shelf life; Italian heirloom variety	52-60 days 3-4" diameter	
Cylindra	Large green tops good for bunching; burgundy cylindrical roots; tender and sweet; Danish heirloom variety	54-56 days 5-6" long 2" diameter	Cercospora leaf spot, scab
Detroit Dark Red	Round, dark red roots with strong tops; uniform and flavorful; good shelf life; adaptable heirloom variety	55-65 days 3" diameter	Downy mildew, late blight
Early Wonder Tall Top	Large, vigorous green tops with red veins for bunching; slightly flattened dark red roots; great for beet greens; very productive heirloom variety	45-55 days (35 days baby leaf, 45 full size leaf) 3-4" diameter	Foliar diseases
Golden or Yellow Detroit	Vibrant orange-yellow roots with green leaves; productive; open-pollinated	55 days 3" diameter	Cercospora leaf spot, scab
Green Top	Bright red roots with flavorful greens; productive; open-pollinated	52 days 3" diameter	
Kestrel	Deep red globe roots; semiglossy, tall tops; sweet and uniform; good shelf life; heat tolerant hybrid	55 days 3-4" diameter	Cercospora leaf spot, downy mildew, powdery mildew, Rhizoctonia root and crown rot
Pacemaker III	Medium, smooth red roots; tall, glossy green tops; flavorful; adaptable to hot and humid regions; heat and drought tolerant hybrid	53 days 3" diameter	Cercospora leaf spot, powdery mildew, Rhizoctonia root and crown rot
Red Ace	Medium, strong green tops with red veins for bunching; uniform red roots; sweet and tender; reliable, adaptable and early maturing; good shelf life; productive hybrid	50-60 days 3" diameter	
Red Cloud	Dark red, large roots; very flavorful and sweet; smooth and vigorous; heat tolerant; hybrid	50-53 days 3" diameter	
Ruby Queen	Dark red, round roots; smooth, tender and sweet; productive and adaptable; open-pollinated	52-55 days 3" diameter	
Solo	Dark red, uniform and round roots; dark green tall tops; sweet; hybrid	50-55 days 3-4" diameter	Cercospora leaf spot, powdery mildew, Rhizoctonia root and crown rot

Variety Name	Description	Days to Harvest & Size	Resistance
Carrot			
Chantenay Red Core	Blocky, broad-shouldered, deep orange roots; adaptable French heirloom Chantenay type variety	65-70 days 5.5" long 2.5" diameter	
Danvers 126	Bright orange, conical, tapered roots; strong tops; sweet and resists splitting; reliable, adaptable, productive and heat tolerant; Danvers type; open-pollinated	70-75 days 6-7" long 2" diameter	
Deep Purple	Dark purple, tapered root; tall, strong, healthy tops; sweet and flavorful; Imperator type hybrid	73 days 7-8" long	
Imperator	Orange-red, coreless roots; sweet, crisp and tender; hybrid	74 days 7-9" long	
Mokum	Slender orange roots; short tops; very flavorful, sweet, and tender; early maturing; adaptable and tolerant to warm weather; early Nantes type hybrid	54 days (36 days baby)	
Nantes Fancy	Dark orange roots with blunt ends; smooth and crisp; good shelf life; productive Nantes type hybrid	68 days 5-8" long	Alternaria leaf spot
Purple Haze	Dark purple, tapered root with bright orange interior; medium-short tops; very flavorful; Imperator type hybrid	73 days 7-8" long	
Scarlet Nantes	Bright red-orange, cylindrical roots; small tops for bunching; fine texture and small core; adaptable French heirloom Nantes type variety	65-70 days 6-8" long 1.5" diameter	
Sugarsnax 54	Dark orange, tapered, slender roots; strong, medium tops; smooth, sweet, and tender; hybrid	68 days 9-10" long	Alternaria blight
Thumbelina	Miniature, round, orange root; tender, very flavorful and productive; good shelf life; open-pollinated	60 days 1-2" globe	
Yellowbunch	Bright yellow roots with strong tops; uniform; Imperator type hybrid	75 days 7.5-8" long	Alternaria blight

Notes: Table varieties selected from recommendations from LSU AgCenter, UF Extension, Texas A&M Extension, and Southeastern U.S. Vegetable Crop Handbook. Variety descriptions compiled from Southern Exposure Seed Exchange, High Mowing Organic Seeds, Johnny's Selected Seeds, Sow True Seed, All-America Selections, Reimer Seeds, Hoss Tools, and Seedway.

Other recommended varieties for Louisiana include:
 Beet: Bohan, Centurion, Chariot, Scarlet Supreme
 Carrot: Apache, Choctaw, Enterprise, Maverick, Navajo, Nelson



Crop Rotation to Prevent Later Problems

- **What is crop rotation?**

- Moving members of a plant *family* to a different location
 - Some less problematic crops can be moved back the second year
 - More pest- or disease-prone crops may not be moved back for 3-4 yrs.

- **Crop rotation can:**

- Prevent disease
- Prevent insect infestation
- Help manage nutrients



Crop Rotation – Why Farmers do it

- Managing plant nutrients
 - Some crops, like corn, are notoriously heavy feeders
 - If grown in the same location every year they will deplete nutrients
 - Leguminous crops form bacteria-filled root nodules that can gather nitrogen from the air and make it available to plant roots
 - Some nitrogen is left in the soil for the following crop
 - Rotating heavy feeders with nitrogen fixers is a sound cultural practice

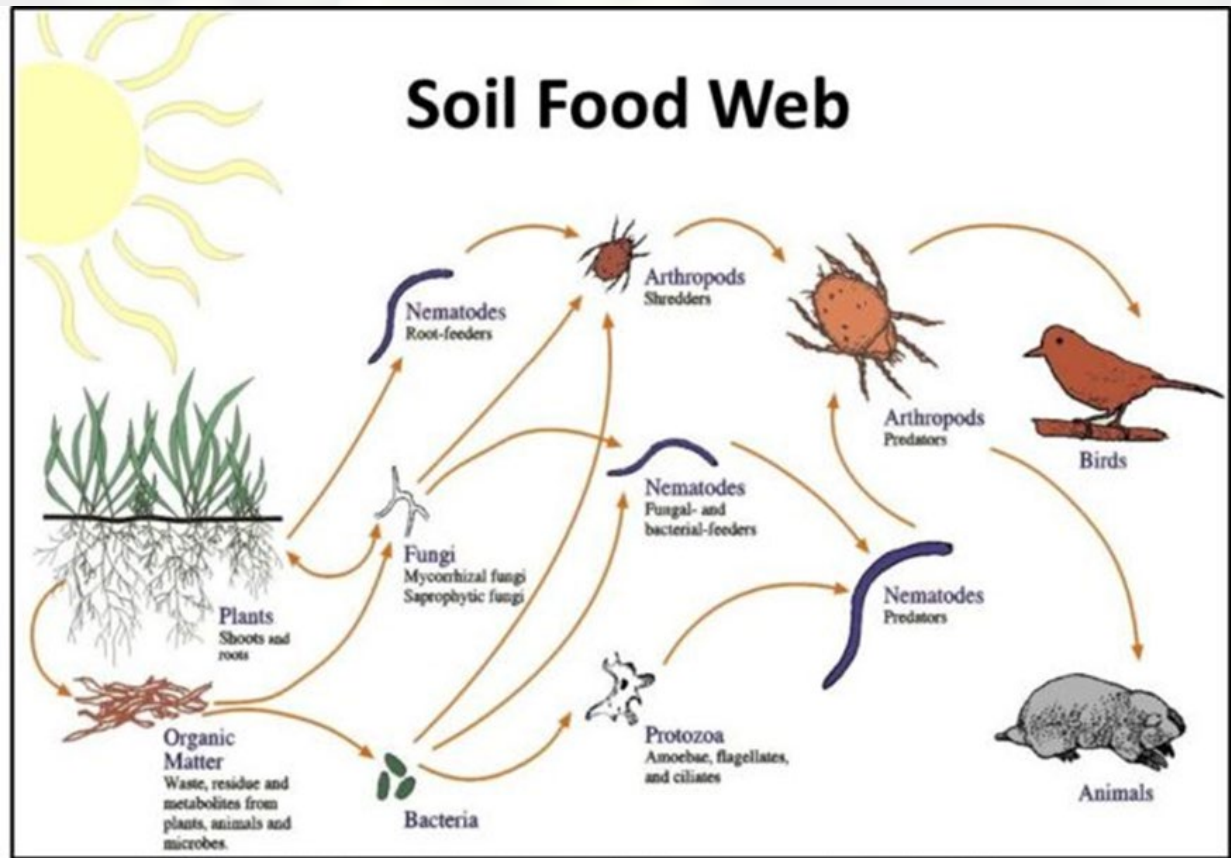


Root nodules on beans.



Crop Rotation – Why Farmers do it

- There's a LOT going on in the soil!
- Different crops attract many different organisms and microorganisms
 - Plant roots leak a little sap into the soil, feeding different microorganisms
 - Some are beneficial soil organisms that are antagonistic towards harmful soil organisms
 - Building a variety of soil microorganisms with crop rotation (growing different crops) is a good cultural practice



Source: The Soil Food Web, USDA Agricultural Research Service 1999



Reasons for Crop Rotation in a School Garden

- Nutrient management
- Building healthy soil (good microorganisms)
- Avoiding insect infestation
- Disease prevention



How Does Crop Rotation Prevent Insect Infestation?

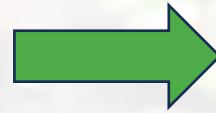
- Many insect life cycles include a pupal stage:
 - Adult insect lays an egg on its preferred host
 - Egg hatches and develops into a hungry larva
 - Larva tunnels through plant while feeding, often destroying a healthy plant
 - Larva exits plant and drops to ground
 - Larva tunnels into soil and morphs into a pupa
 - Pupa overwinters in the ground as it develops into an adult
 - Adult insect emerges from the earth in spring
 - Adult lays eggs on its preferred host plant; cycle starts over
- Is the same host plant there again this year? **Reinfestation!**



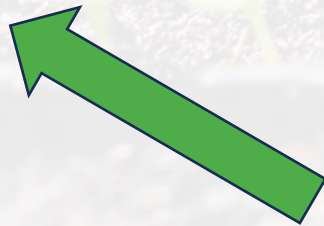
Tomato hornworm pupa.



Squash Vine Borer



Pupa underground.



If squash is there next year, the cycle starts over.
Rotate to another row.



How does crop rotation prevent disease?

- Many of our favorite crops are very disease-prone
 - Example: Tomatoes have too many diseases to list (but we cherish them anyway)
 - Some diseases are soilborne
 - After infection, you can pull the dead plant, but fungal or bacterial disease spores remain behind in the soil!
 - Some diseases are fungal or bacterial foliage diseases
 - Infected leaves fall to the ground, spores remain in the soil

It can take 3-5 YEARS for spores to die out

- Tomatoes are members of the *Solanaceae* family (also called nightshades)
 - So are eggplants, potatoes, and peppers
 - They can ALL be infected with the same diseases



Fusarium Wilt



Tomato



Pepper



Early Blight



Potato



Tomato



Following winter potatoes with spring tomatoes
(both *Solanaceae*)
is a plan for potential disaster.

To avoid future problems, **make a plan** for rotation
(not following with another *Solanaceae*)
for 3 years



Include a CROP ROTATION plan in your planning

- Knowing which *FAMILY* a crop fits into is essential to planning a crop rotation scheme

YEAR 1	YEAR 2	YEAR 3	YEAR 4
PLOT 1 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 1 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 1 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 1 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery
PLOT 2 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 2 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 2 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 2 Broad Beans French Beans Runner Beans Peas Sweet Potato
PLOT 3 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 3 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 3 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 3 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket
PLOT 4 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 4 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 4 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 4 Potatoes Tomatoes Chilli Peppers Sweet Peppers



Which *Families* are your favorite crops in?



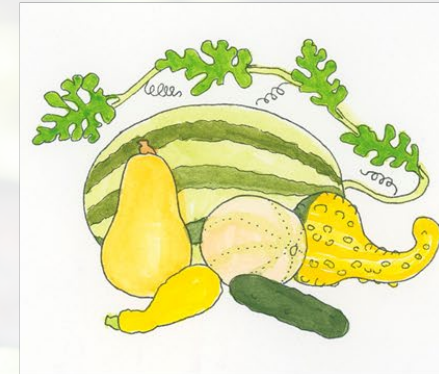
Solanaceae
Nightshades



Brassicaceae
Brassica or Cole



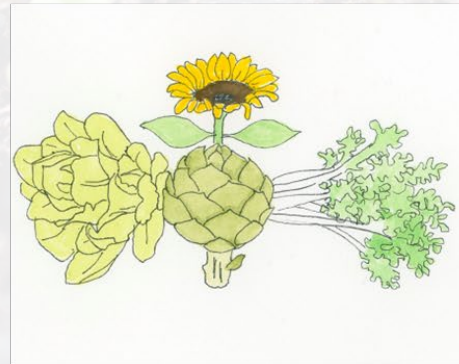
Chenopodiaceae
Goosefoot



Cucurbitaceae
Cucurbits



Apiaceae
Carrot



Asteraceae
Aster or Sunflower



Fabaceae
Pea or Legume



Poaceae
Cereal or Grain



Seeds of a *Family* Look Alike



Brassica
Brassicaceae

Broccoli
Radish
Cabbage
Collards
Cauliflower



Cucurbit
Cucurbitaceae

Melon
Cucumber
Squash
Pumpkin



Nightshade
Solanaceae

Tomato
Pepper
Potato
Eggplant



Legumes
Fabaceae

Beans
Peas
Lentils
Field peas



Their seeds look alike... How about the flowers? And seed receptacles?

Cucurbit



Nightshade



Brassica



Legume



Louisiana Sustainable Planting Guide for School and Home Gardens

Crop	LA Outdoor Planting Dates ^a		Recommended Planting Method ^b	Seed Depth (inch)	Plant Spacing (inches)		Days to Harvest ^c	Plant Family
	North Zone 8	South Zone 9			In-row	Between Rows		
Arugula	Sept-Mar	Sept-Mar	DS	1/4	3-4	10	35-60	<i>Brassicaceae</i>
Beet	Feb-Mar Sept 15-Nov 15	Jan 15-Mar 15 Aug 15-Nov 15	DS	1/8-1/4	2-4	10-12 double-set	50-70	<i>Chenopodiaceae</i>
Bean, Lima (Butter)	April-July 25	Mar-Aug 15	DS	1	2-4	Bush: 18 Pole: 36-48	60-80	<i>Fabaceae</i>
Bean, Snap (Bush)	April-May 15 Aug-Sept	Feb-May 15 Aug-Sept	DS	1	2-4	18 double-set	45-60	<i>Fabaceae</i>
Bean, Snap (Pole)	April-May 15 Aug-Sept	Feb-May 15 Aug-Sept	DS	1	6-12	36-48	50-70	<i>Fabaceae</i>
Blueberries	Bare root: Nov-Feb Container: Nov-May		T	NA	6-8 feet	10-12 feet	2-3 years	<i>Ericaceae</i>
Broccoli	Feb-Mar 15 Aug-Oct	Jan 15-Mar 15 Aug-Oct	T (5-7)	1/8-1/4	10-24	12-24 double-set	75-100 (50-70)	<i>Brassicaceae</i>
Brussels Sprouts	Aug-Feb	Sept-Feb	T (5-7)	1/8	9-16	24	90	<i>Brassicaceae</i>
Cabbage	Jan 15-Mar 15 Aug-Nov	Jan 15-Mar 15 Aug 15-Nov	T (6) or DS	1/4-1/2	9-18	12-36 double-set	85-110 (70-90)	<i>Brassicaceae</i>
Cabbage, Chinese	Jan 15-Mar 15 Aug-Oct		T (6) or DS	1/4-1/2	6-12 full size 2-4 baby	18 full size 3 baby double-set	70-100 (40-70)	<i>Brassicaceae</i>
Cantaloupe^d	April-July	Mar 15-Aug 15	DS	1/2	12-36 or hills	60-72	85-110 (70-90)	<i>Cucurbitaceae</i>
Carrot	Jan 15-Mar Aug 15-Oct 15	Jan 15-Mar 15 Aug 15-Nov 15	DS	1/8-1/4	1-3	10-12 double-set	70-90	<i>Apiaceae</i>
Cauliflower	Feb-Mar 15 July 15-Oct	Jan 15-Feb 15 July-Oct 15	T (5-7)	1/8-1/4	18-24	24-36	75-90 (50-70)	<i>Brassicaceae</i>
Celery	Aug-Feb	Sept-Feb	T (8-10)	1/8	6-12	18	210	<i>Apiaceae</i>
Citrus, Orange and Grapefruit	Jan-Mar		T	NA	30-40 feet diameter		3-4 years	<i>Rutaceae</i>
Citrus, Satsuma, Tangelo, Tangerine, Mandarin	Jan-Mar		T	NA	20-30 feet diameter		3-4 years	<i>Rutaceae</i>



Make a Rotation Plan for fall crops

Year 1

Bed1

Year 2

Bed 1

Year 3

Bed 1

- Use the Planting Guide **Plant Family** column
- Do not move a *family* back to the same spot in the bed for 3 years



Solonaceae

Tomato
Pepper
Eggplant
Potato



Brassicaceae

Collards
Turnips
Broccoli
Cauliflower
Radish



Apiaceae

Carrots
Celery
Parsley
Fennel



Asteraceae

Lettuce
Artichoke
Sunflower



Cucurbitaceae

Cucurbits
Squash
Pumpkins
Cukes



Fabaceae

Peas
Beans
Cowpeas

Bed 2

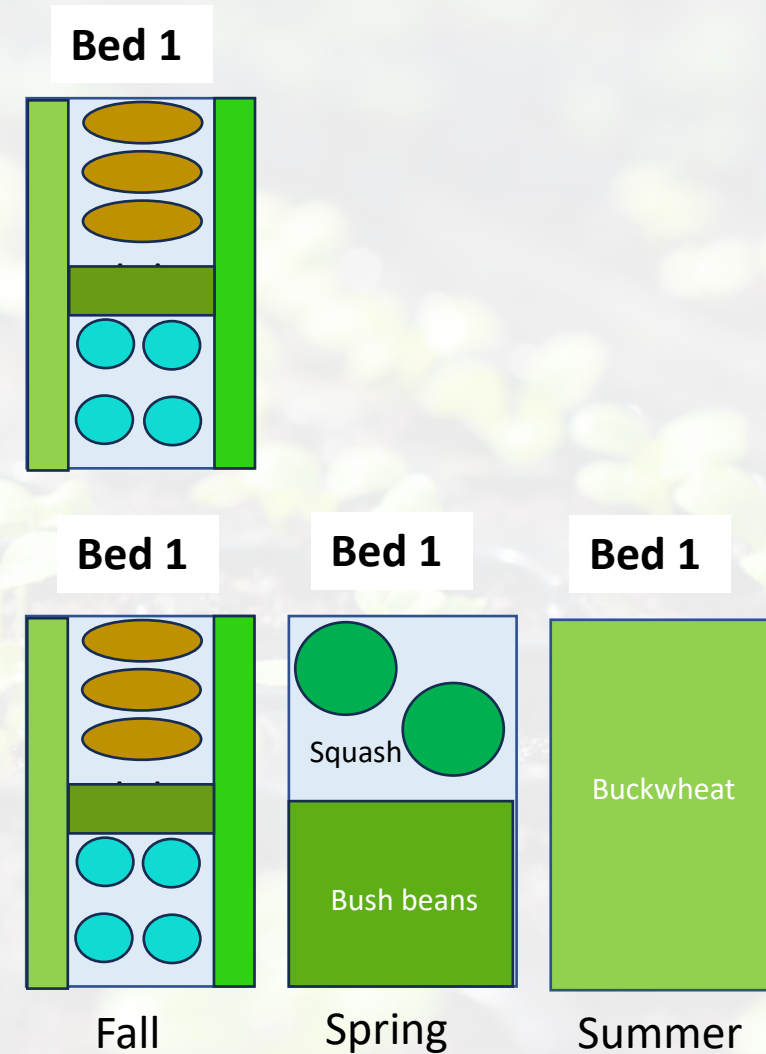
Bed 2

Bed 2



Rotation Plans

- You may want to plan just for fall, when you're using the garden for lessons.
- You may want to plan for year-round, adding some quick spring crops like squash or green beans, and planning for summer, when the bed will be cleaned and covered with plastic or growing a cover crop like buckwheat or cowpeas.



- You could choose to plant only one family in the entire bed, switching to different families the following 2 years.
- Choose crops kids like to grow and are easy to prepare and eat. There are INFINITE possibilities for rotation now that you know the plant families.

YEAR 1	YEAR 2	YEAR 3	YEAR 4
PLOT 1 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 1 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 1 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 1 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery
PLOT 2 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 2 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 2 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 2 Broad Beans French Beans Runner Beans Peas Sweet Potato
PLOT 3 Potatoes Tomatoes Chilli Peppers Sweet Peppers	PLOT 3 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 3 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 3 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket
PLOT 4 Beetroot Carrots Garlic Leeks Onions Parsnips Salsify Celeriac Celery	PLOT 4 Broad Beans French Beans Runner Beans Peas Sweet Potato	PLOT 4 Broccoli Brussels Sprouts Cabbage Calabrese Cauliflower Kale Kohl-Rabi Mustard Swede Turnips Chard Spinach Rocket	PLOT 4 Potatoes Tomatoes Chilli Peppers Sweet Peppers



For this exercise, plan for 3 years of fall crops, being mindful not to place plants of the same family where they were grown before



Which *Families* are your favorite crops in?



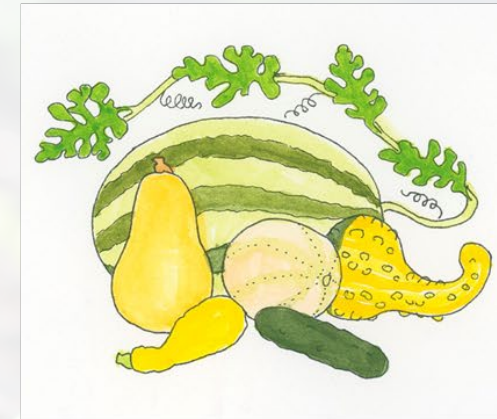
Solanaceae
Nightshades



Brassicaceae
Brassica or Cole



Chenopodiaceae
Goosefoot



Cucurbitaceae
Cucurbits



Apiaceae
Carrot



Asteraceae
Aster or Sunflower



Fabaceae
Pea or Legume

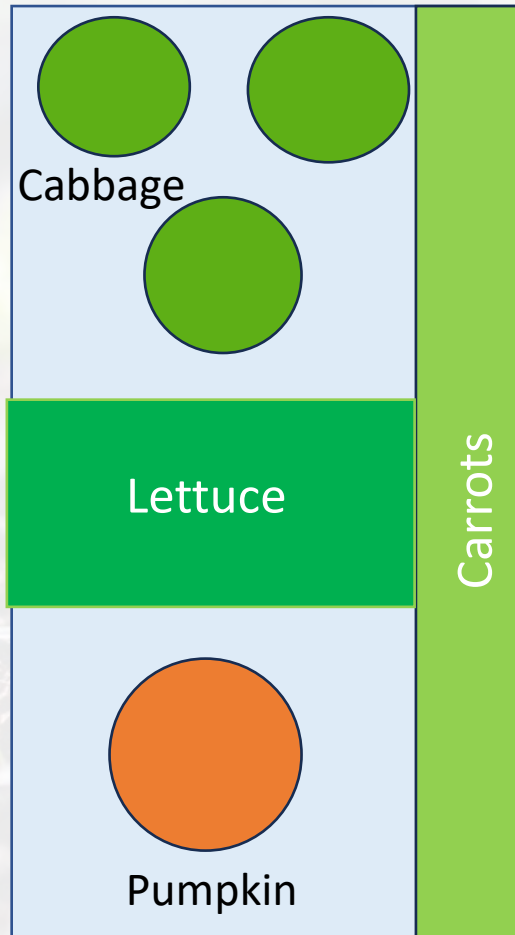


Poaceae
Cereal or Grain

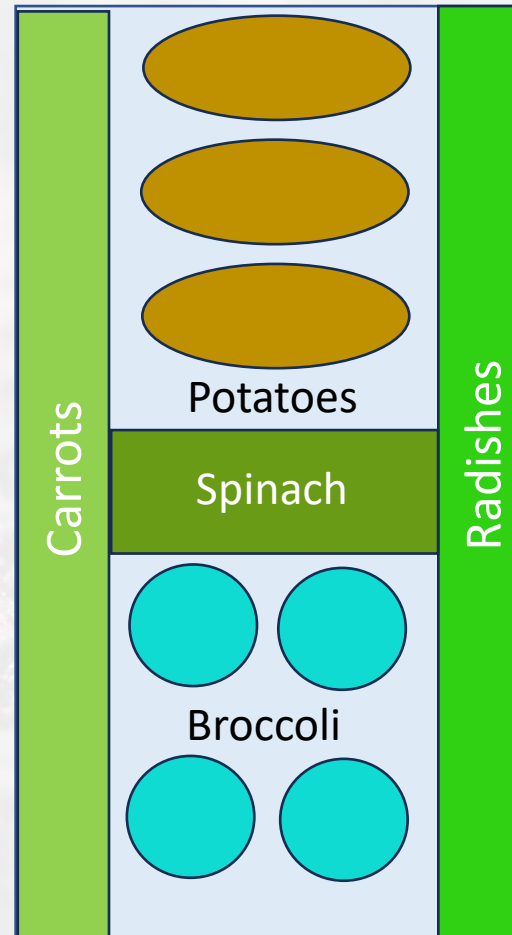


My Bed 1

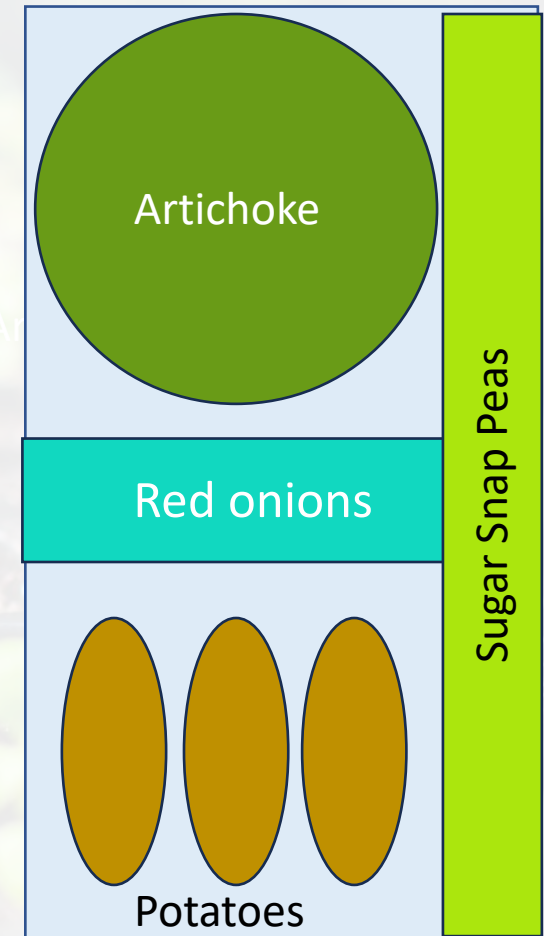
Year 1



Year 2



Year 3



Cabbage, broccoli and radishes are *Brassicas*. Artichoke and lettuce are *Asteraceae*. They cannot follow each other and must be rotated to another area.



Questions?

dcummins@agcenter.lsu.edu



**GROWING NEW
CONNECTIONS**

2023 Louisiana Farm to School Conference